Cost Control

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AWS Cost Allocation Tags

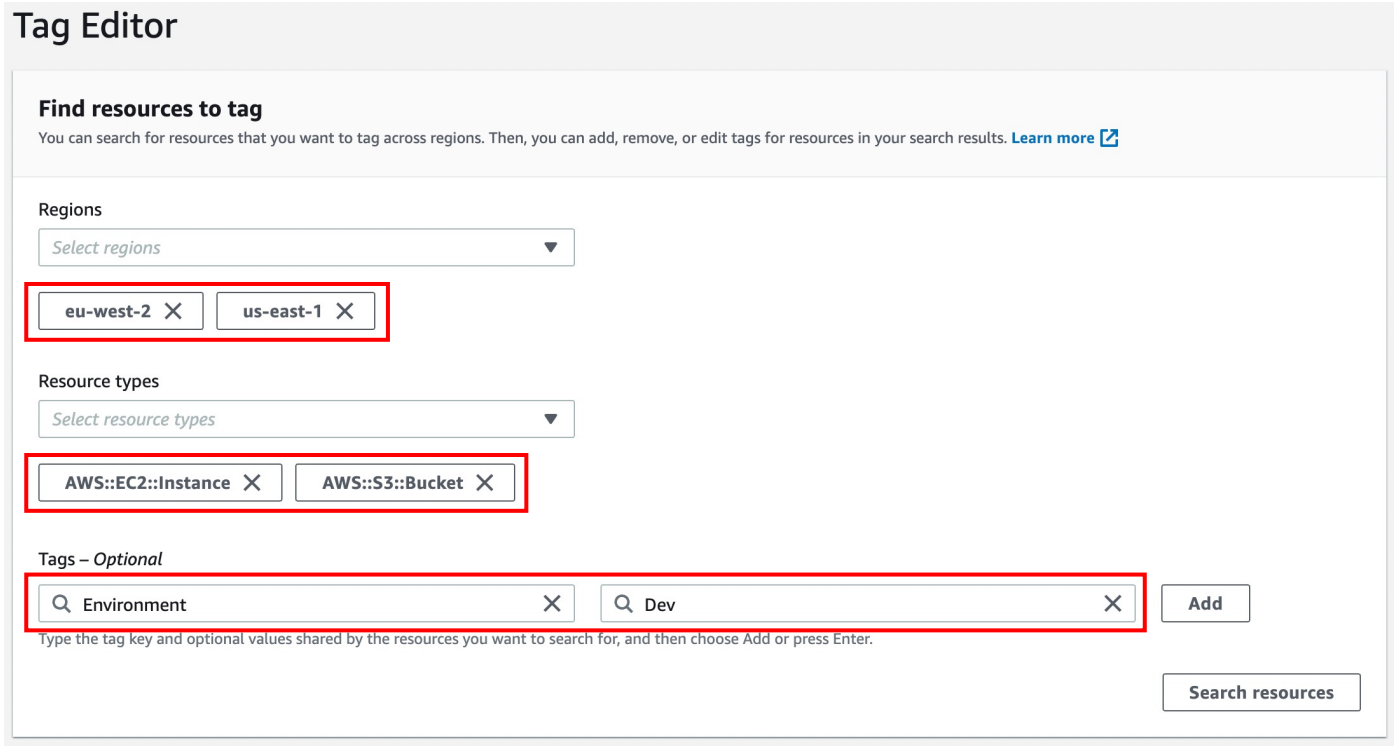
- With Tags we can track resources that relate to each other
- With Cost Allocation Tags we can enable detailed costing reports
- **Just like Tags, but they show up as columns in Reports**
- AWS Generated Cost Allocation Tags
  - Automatically applied to the resource you create
  - Starts with Prefix **aws:** **(e.g. aws: createdBy)**
  - They're not applied to resources created before the activation
- User tags
  - Defined by the user
  - Starts with Prefix **user:**
- **Cost Allocation Tags just appear in the Billing Console**
- Takes up to 24 hours for the tags to show up in the report

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AWS Tag Editor

- Allows you to manage tags of multiple resources at once
- You can add/update/delete tags
- Search tagged/untagged resources in all AWS Regions



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Trusted Advisor



- No need to install anything – high level AWS account assessment
- Analyze your AWS accounts and provides recommendation:
  - Cost Optimization
  - Performance
  - Security
  - Fault Tolerance
  - Service Limits
  - Operational Excellence
- Core Checks and recommendations – all customers
- Can enable weekly email notification from the console
- Full Trusted Advisor – Available for Business & Enterprise support plans
  - Ability to set CloudWatch alarms when reaching limits
  - Programmatic Access using AWS Support API

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AWS Support Plans

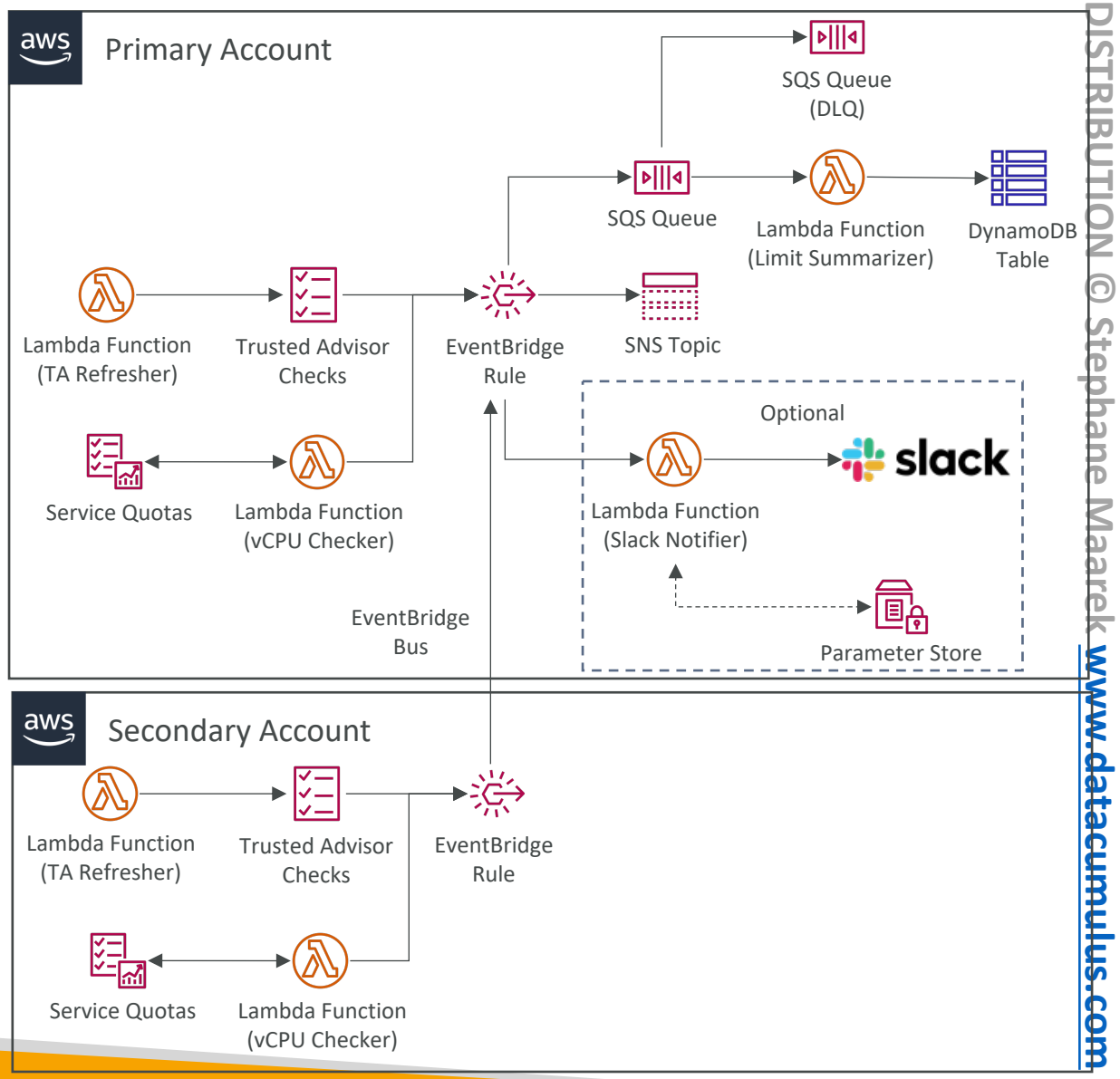
	Basic Support	Developer	Business	Enterprise
	*included for all AWS customers and **free***	*Recommended if you are experimenting or testing in AWS.*	*Recommended if you have production workloads in AWS.*	*Recommended if you have business and/or mission critical workloads in AWS.*
AWS Trusted Advisor Best Practice Checks	7 Core checks	7 Core checks	**Full set of checks + Programmatic Access using AWS Support API**	**Full set of checks + Programmatic Access using AWS Support API**
Enhanced Technical Support	24x7 access to customer service, documentation, whitepapers, and support forums.	Business hours email access to Cloud Support Associates Unlimited cases / 1 primary contact	24x7 phone, email, and chat access to Cloud Support Engineers Unlimited cases / unlimited contacts (IAM supported)	24x7 phone, email, and chat access to Cloud Support Engineers Unlimited cases / unlimited contacts (IAM supported)
Case Severity / Response Times		General guidance: < 24 business hours** System impaired: < 12 business hours**	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour Business-critical system down: < 15 minutes

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Trusted Advisor – Good to know

- Can check if an S3 bucket is made public
  - But cannot check for S3 objects that are public inside of your bucket!
  - Use Amazon EventBridge / S3 Events instead / AWS Config Rules
- Service Limits
  - **Limits can only be monitored** in Trusted Advisor (cannot be changed)
  - Cases must be created manually in **AWS Support Centre** to increase limits
  - OR use the **AWS Service Quotas** service

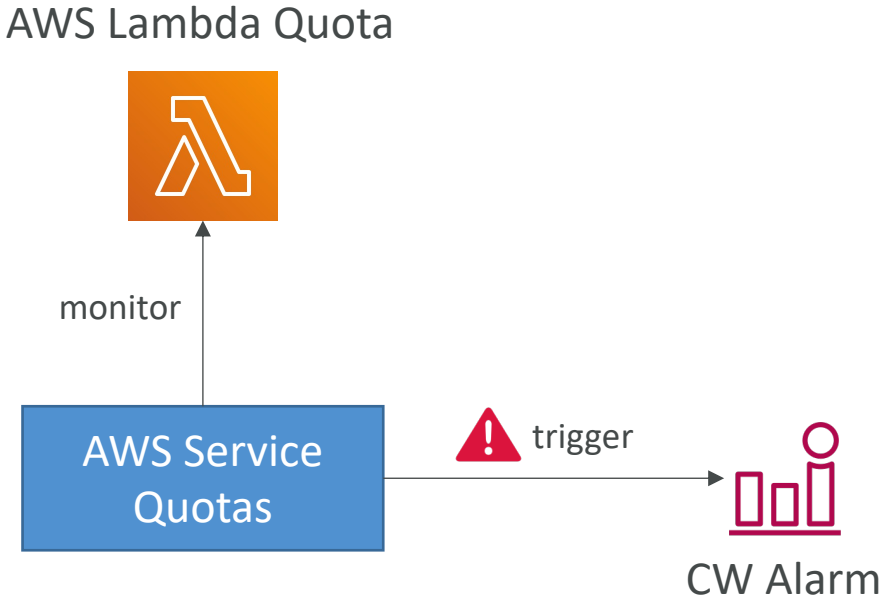


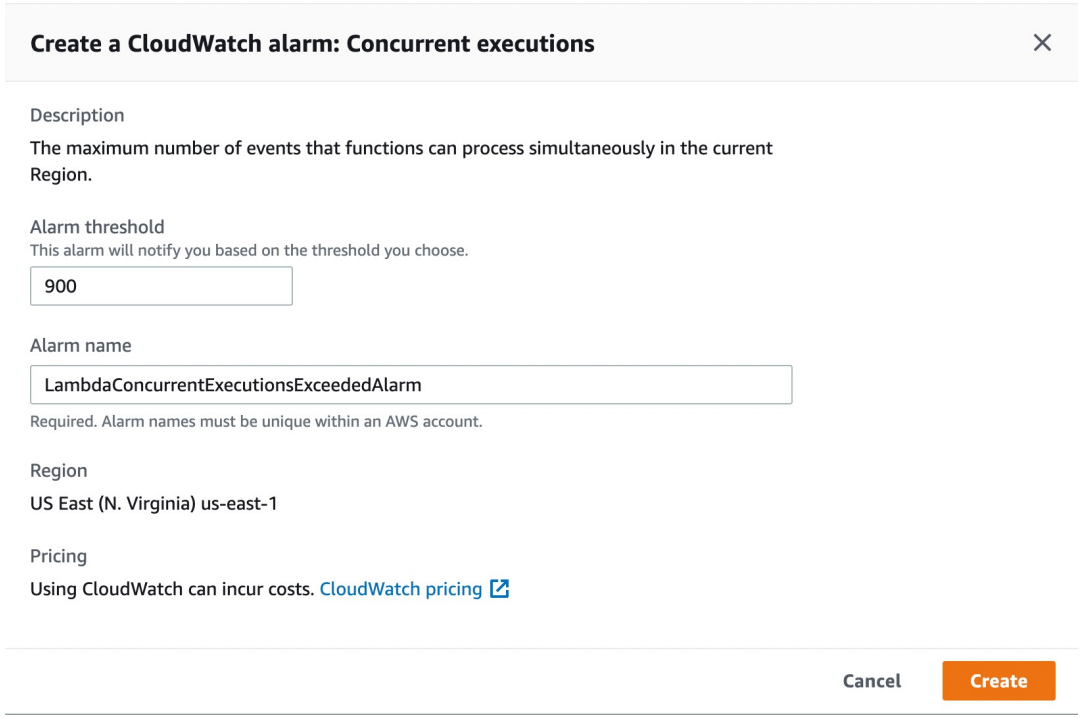
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AWS Service Quotas

- Notify you when you're close to a service quota value threshold
- Create CloudWatch Alarms on the Service Quotas console
- Example: Lambda concurrent executions
- Helps you know if you need to request a quota increase or shutdown resources before limit is reached





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EC2 Instance Launch Types

- **On Demand Instances:** short workload, predictable pricing, reliable
- **Spot Instances:** short workloads, for cheap, can lose instances (not reliable)
- **Reserved:** (MINIMUM 1 year)
  - **Reserved Instances:** long workloads
  - **Convertible Reserved Instances:** long workloads with flexible instances
- **Dedicated Instances:** no other customers will share your hardware
- **Dedicated Hosts:** book an entire physical server, control instance placement
  - Great for software licenses that operate at the core, or socket level
  - Can define **host affinity** so that instance reboots are kept on the same host

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AWS Savings Plan



- New pricing model to get a discount based on long-term usage
- Commit to a certain type of usage: ex $10 per hour for 1 to 3 years
- Any usage beyond the savings plan is billed at the on-demand price

- **EC2 Instance Savings plan** (up to 72% - same discount as Standard RIs)
  - Select instance family (e.g. M5, C5…), and locked to a specific region
  - Flexible across size (m5.large to m5.4xlarge), OS (Windows to Linux), tenancy (dedicated or default)
- **Compute Savings plan** (up to 66% - same discount as Convertible RIs)
  - Ability to move between instance family (move from C5 to M5), region (Ireland to US), compute type (EC2, Fargate, Lambda), OS & tenancy
- **SageMaker Savings plan** (up to 64% off)

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S3 Storage Classes

- Amazon S3 Standard - General Purpose
- Amazon S3 Standard-Infrequent Access (IA)
- Amazon S3 One Zone-Infrequent Access
- Amazon S3 Glacier Instant Retrieval
- Amazon S3 Glacier Flexible Retrieval
- Amazon S3 Glacier Deep Archive
- Amazon S3 Intelligent Tiering

- Can move between classes manually or using S3 Lifecycle configurations

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S3 – Other Cost Savings

- **S3 Lifecycle Rules:** transition objects between tiers
- **Compress objects** to save space
- **S3 Requester Pays:**
  - In general, bucket owners pay for all Amazon S3 storage and data transfer costs associated with their bucket
  - With Requester Pays buckets, the requester instead of the bucket owner pays the cost of the request and the data download from the bucket
  - The bucket owner always pays the cost of storing data
  - Helpful when you want to share large datasets with other accounts
  - **If an IAM role is assumed, the owner account of that role pays for the request**

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S3 Durability and Availability

- Durability:
  - High durability (99.999999999%, 11 9's) of objects across multiple AZ
  - If you store 10,000,000 objects with Amazon S3, you can on average expect to incur a loss of a single object once every 10,000 years
  - Same for all storage classes

- Availability:
  - Measures how readily available a service is
  - Varies depending on storage class
  - Example: S3 standard has 99.99% availability = not available 53 minutes a year

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S3 Standard – General Purpose



- 99.99% Availability
- Used for frequently accessed data
- Low latency and high throughput
- Sustain 2 concurrent facility failures

- Use Cases: Big Data analytics, mobile & gaming applications, content distribution…

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S3 Storage Classes – Infrequent Access

- For data that is less frequently accessed, but requires rapid access when needed
- Lower cost than S3 Standard

- Amazon S3 Standard-Infrequent Access (S3 Standard-IA)
  - 99.9% Availability
  - Use cases: Disaster Recovery, backups



- Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)
  - High durability (99.999999999%) in a single AZ; data lost when AZ is destroyed
  - 99.5% Availability
  - Use Cases: Storing secondary backup copies of on-premise data, or data you can recreate



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Amazon S3 Glacier Storage Classes

- Low-cost object storage meant for archiving / backup
- Pricing: price for storage + object retrieval cost

- Amazon S3 Glacier Instant Retrieval
  - Millisecond retrieval, great for data accessed once a quarter
  - Minimum storage duration of 90 days
- Amazon S3 Glacier Flexible Retrieval (formerly Amazon S3 Glacier):
  - Expedited (1 to 5 minutes), Standard (3 to 5 hours), Bulk (5 to 12 hours) – free
  - Minimum storage duration of 90 days
- Amazon S3 Glacier Deep Archive – for long term storage:
  - Standard (12 hours), Bulk (48 hours)
  - Minimum storage duration of 180 days



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S3 Intelligent-Tiering



- Small monthly monitoring and auto-tiering fee
- Moves objects automatically between Access Tiers based on usage
- There are no retrieval charges in S3 Intelligent-Tiering

- *Frequent Access tier (automatic):* default tier
- *Infrequent Access tier (automatic)*: objects not accessed for 30 days
- *Archive Instant Access tier (automatic):* objects not accessed for 90 days
- *Archive Access tier (optional):* configurable from 90 days to 700+ days
- *Deep Archive Access tier (optional):* config. from 180 days to 700+ days

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S3 Storage Classes Comparison

	Standard	Intelligent-Tiering	Standard-IA	One Zone-IA	Glacier Instant Retrieval	Glacier Flexible Retrieval	Glacier Deep Archive
Durability	99.999999999% == (11 9's)						
Availability	99.99%	99.9%	99.9%	99.5%	99.9%	99.99%	99.99%
Availability SLA	99.9%	99%	99%	99%	99%	99.9%	99.9%
Availability Zones	>= 3	>= 3	>= 3	1	>= 3	>= 3	>= 3
Min. Storage Duration Charge	None	None	30 Days	30 Days	90 Days	90 Days	180 Days
Min. Billable Object Size	None	None	128 KB	128 KB	128 KB	40 KB	40 KB
Retrieval Fee	None	None	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved	Per GB retrieved

https://aws.amazon.com/s3/storage-classes/

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S3 Storage Classes – Price Comparison Example: us-east- l

	Standard	Intelligent-Tiering	Standard-IA	One Zone-IA	Glacier Instant Retrieval	Glacier Flexible Retrieval	Glacier Deep Archive
Storage Cost (per GB per month)	$0.023	$0.0025 - $0.023	$0.0125	$0.01	$0.004	$0.0036	$0.00099
Retrieval Cost (per 1000 request)	**GET:** $0.0004 **POST:** $0.005	**GET:** $0.0004 **POST:** $0.005	**GET:** $0.001 **POST:** $0.01	**GET:** $0.001 **POST:** $0.01	**GET:** $0.01 **POST:** $0.02	**GET:** $0.0004 **POST:** $0.03 **Expedited:** $10 **Standard:** $0.05 **Bulk:** free	**GET:** $0.0004 **POST:** $0.05 **Standard:** $0.10 **Bulk:** $0.025
Retrieval Time	Instantaneous					**Expedited** (1 – 5 mins) **Standard** (3 – 5 hours) Bulk (5 – 12 hours)	**Standard** (12 hours) **Bulk** (48 hours)
Monitoring Cost (per 1000 objects)		$0.0025					

https://aws.amazon.com/s3/pricing/

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AWS Budgets



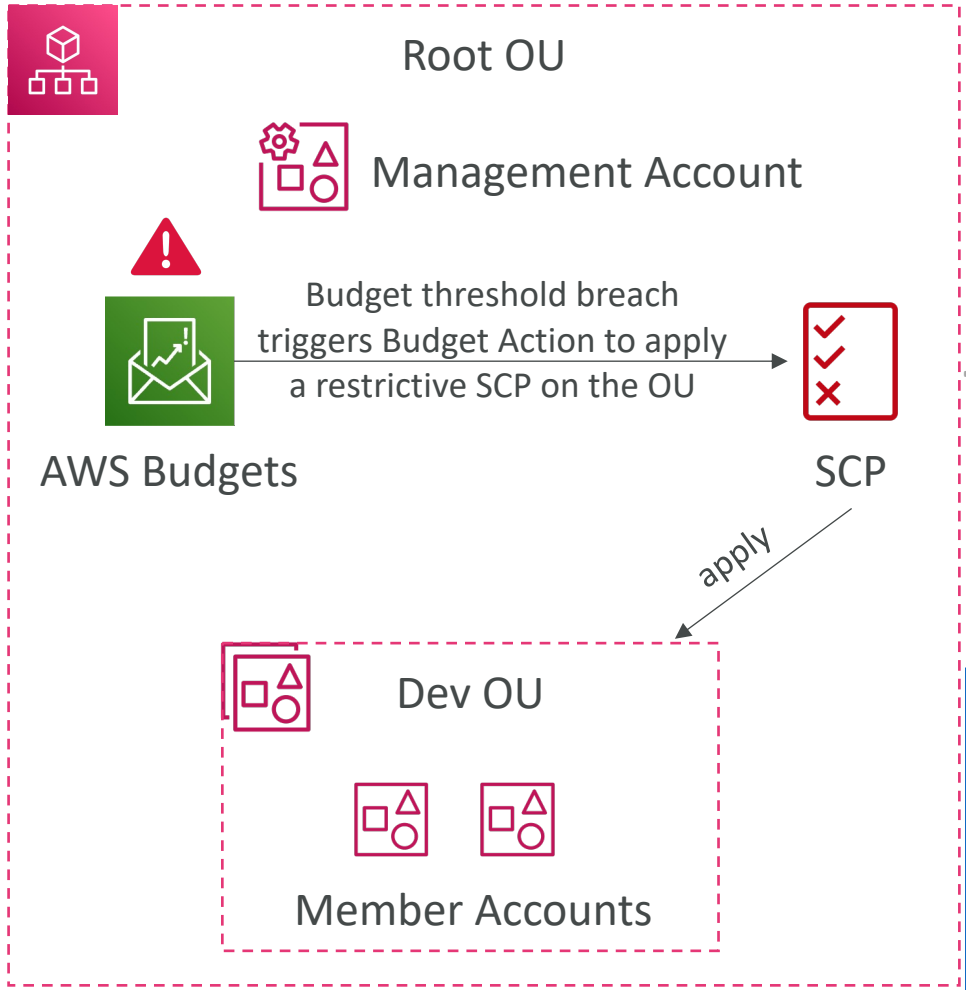
- Create budget and **send alarms when costs exceeds the budget**
- 4 types of budgets: Usage, Cost, Reservation, Savings Plans
- For Reserved Instances (RI)
  - Track utilization
  - Supports EC2, ElastiCache, RDS, Redshift
- Up to 5 SNS notifications per budget
- Can filter by: Service, Linked Account, Tag, Purchase Option, Instance Type, Region, Availability Zone, API Operation, etc…
- Same options as AWS Cost Explorer!
- 2 budgets are free, then $0.02/day/budget

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Budget Actions

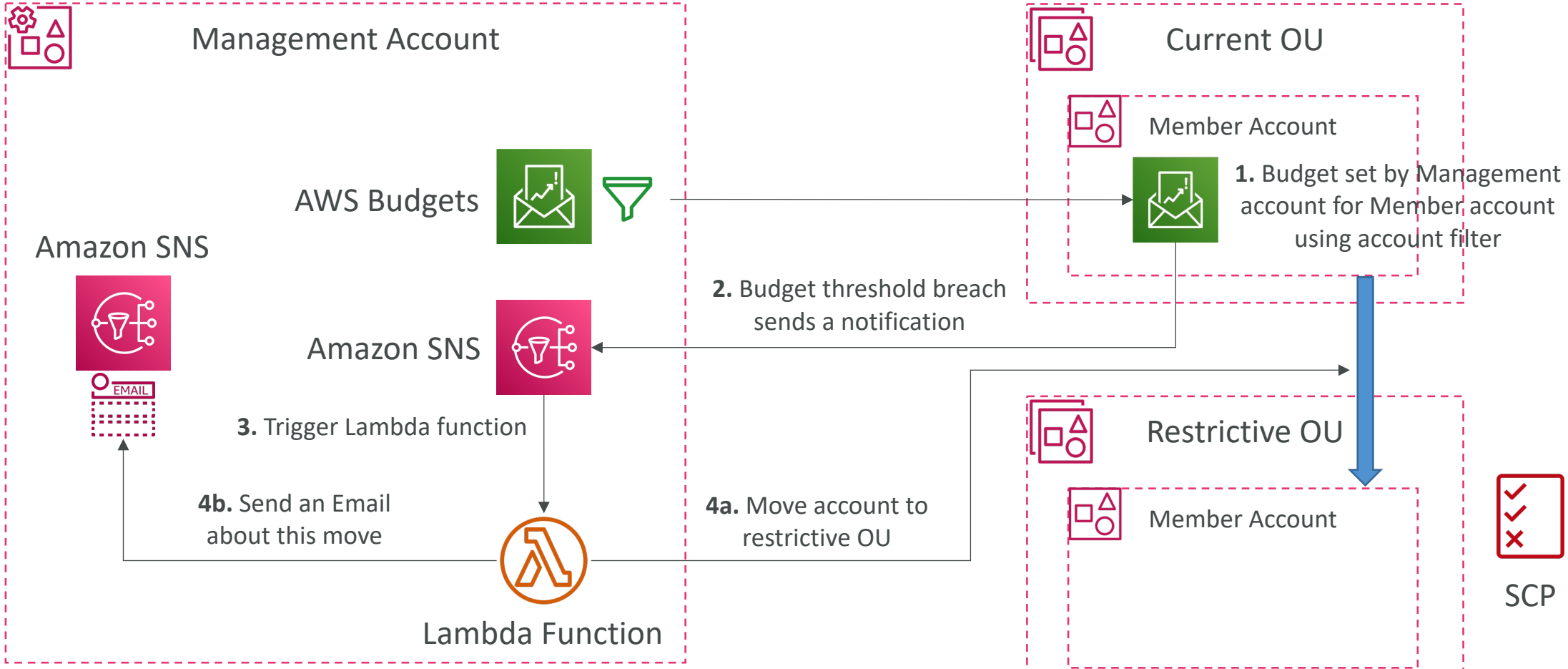
- Run actions on your behalf when a budget exceeds a certain cost or usage threshold
- Supports 3 action types:
  - Applying an IAM Policy to a user, group, or IAM role
  - Applying Service Control Policy (SCP) to an OU
  - Stop EC2 or RDS Instances
- Actions can be executed automatically or require a workflow approval process
- Reduces unintentional overspending in your account



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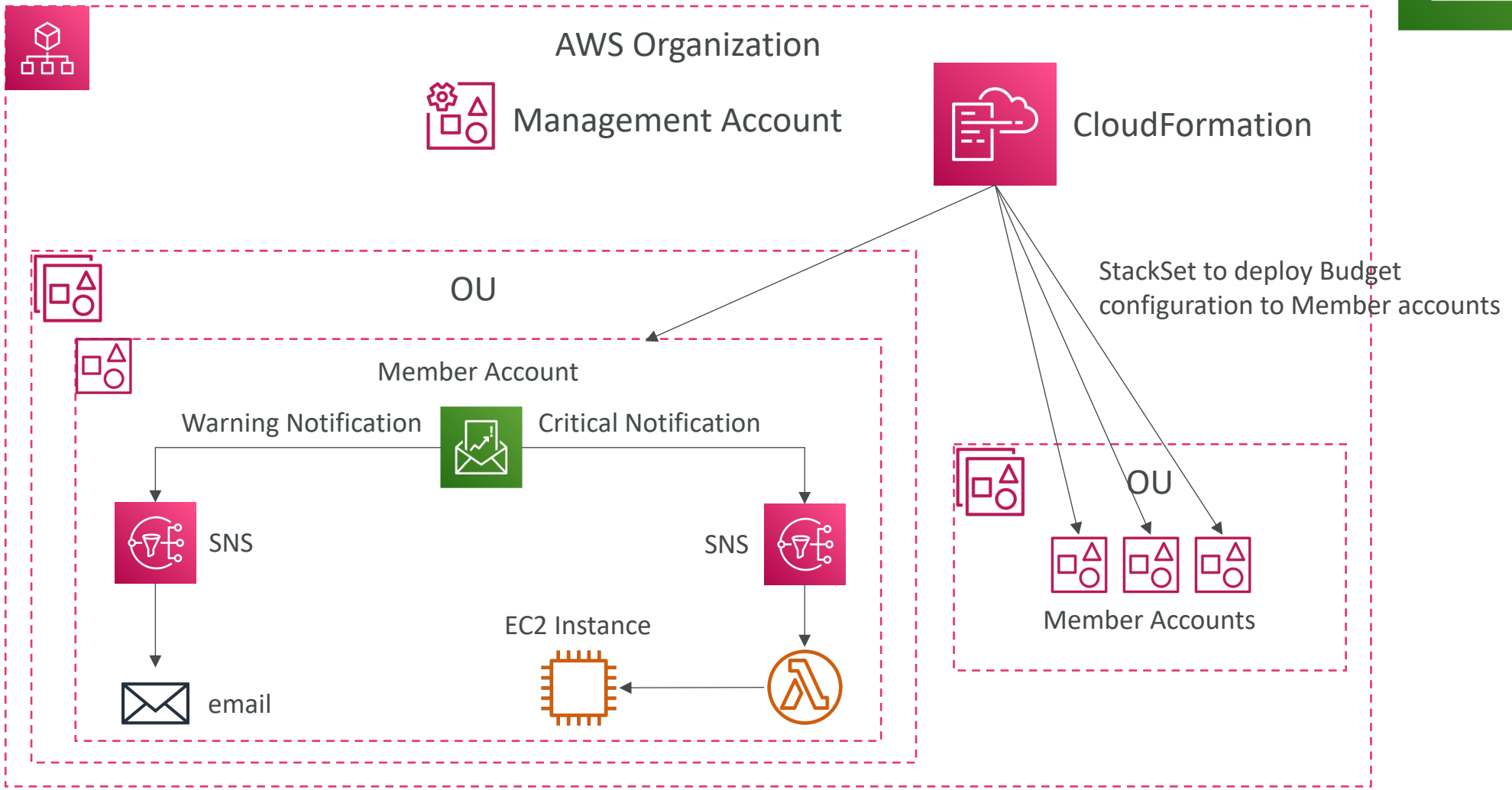
Centralized Budget Management



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Decentralized Budget Management



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Cost Explorer



- Visualize, understand, and manage your AWS costs and usage over time
- Create custom reports that analyze cost and usage data.
- Analyze your data at a high level: total costs and usage across all accounts
- Or Monthly, hourly, resource level granularity
- Choose an optimal **Savings Plan** (to lower prices on your bill)
- **Forecast usage up to 12 months based on previous usage**

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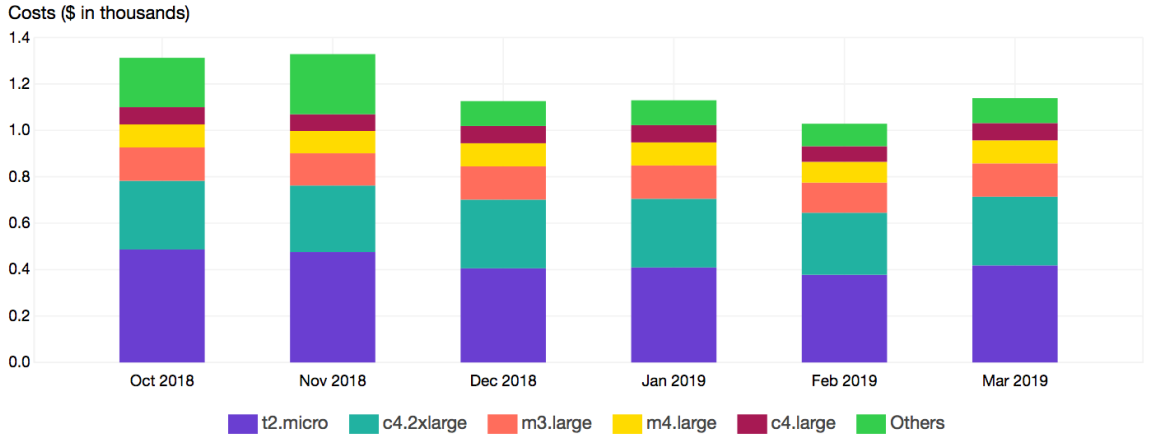
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Cost Explorer – Example

Monthly costs by service

Last 6 Months | Monthly | Stack

Group by: Instance Type | Service | Linked Account | Region | Usage Type | Tag | API Operation | Availability Zone | More



Download CSV

Instance Type	Oct 1, 2018	Nov 1, 2018	Dec 1, 2018	Jan 1, 2019
Total cost ($)	1,312.71	1,328.54	1,125.99	1,129.65
t2.micro ($)	486.75	475.89	405.63	409.27
c4.2xlarge ($)	296.11	286.56	296.11	296.11

FILTERS CLEAR ALL

Service: Include only — EC2-Instances ✖ 1

Linked Account: Include all

Region: Include all

Instance Type: Include all

Usage Type: Include all

Usage Type Group: Include all

Tag: Include All

API Operation: Include all

Charge Type: Include all

More filters

ADVANCED OPTIONS

Show costs as

Unblended costs

Include costs related to

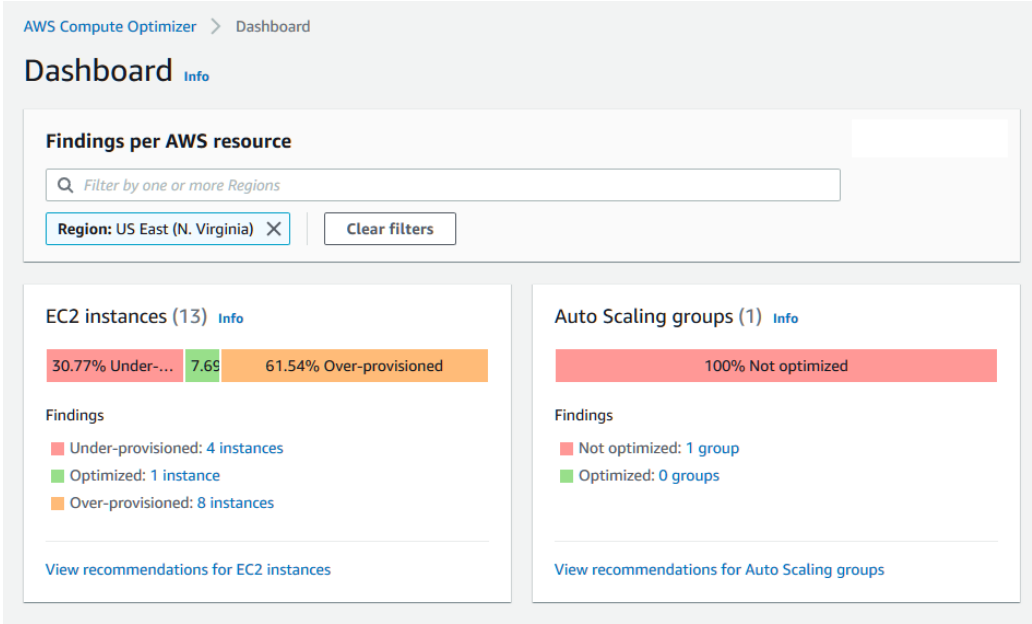
☐ Show only untagged resources

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AWS Compute Optimizer

- **Reduce costs** and **improve performance by** recommending optimal AWS resources for your workloads
- Helps you choose optimal configurations and right-size your workloads (over/under provisioned)
- Uses Machine Learning to analyze your **resources' configurations** and their **utilization CloudWatch metrics**
- Supported resources
  - EC2 instances
  - EC2 Auto Scaling Groups
  - EBS volumes
  - Lambda functions
- Lower your costs by up to 25%
- Recommendations can be exported to S3

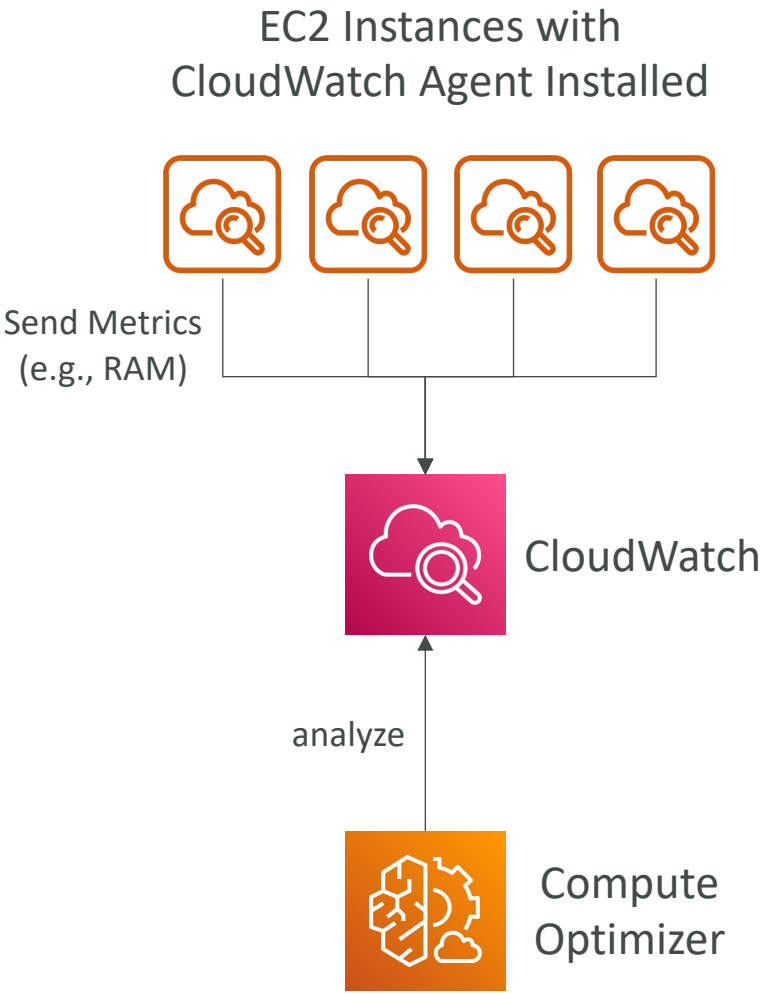


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Compute Optimizer – CloudWatch Agent

- Needed to analyze Memory Utilization
- Not needed for CPU, NetworkIn/Out, DiskReadOps, DiskWriteOps, …



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Reserved Instances

- Reserved Instances in an AWS Organization
  - all accounts share the Reserved Instances and Savings Plan
  - The payer account (Management account) of an organization can turn off Reserved Instance (RI) discount and Savings Plans discount sharing for any accounts in that organization, including the payer account

- Renewal of Reserved Instances
  - You can queue (schedule and reserve ahead of time) your reserved instances
  - To renew a RI, just queue an RI purchase whenever the previous one expires

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